NIRAJ CHARPE

Software Engineer at HCLTech

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PROFESSIONAL SUMMARY

Software engineer at HCLTech, building practical software across backend development, databases, web technologies, and AI. Completed Class 12 and developed programming foundations through professional experience and self-directed project work. Focused on backend engineering, database systems, and enterprise technologies.

TECHNICAL SKILLS

Used in projects: Python, JavaScript, TypeScript, React, Next.js, Tailwind CSS, FastAPI, Flask, Django, Node.js/Express, SQLite and PostgreSQL basics, Google Gemini API, PyTorch, Docker, Git, GitHub, Vercel, Monaco Editor.
Practising / learning: Java, C++, SQL Server and MySQL fundamentals, data structures and algorithms, system design, testing, CI/CD, SAP ABAP fundamentals, cloud fundamentals.
Familiar with fundamentals: C, Linux and shell basics, REST API design, web security basics, Agile/Scrum, technical writing.

FEATURED PROJECTS

CareerLens AI - Next.js, FastAPI, TypeScript, Gemini API

Resume analysis platform that parses PDF/DOCX files and maps skills to career paths with a learning plan.

- Client and API validation; PyMuPDF and python-docx extraction
- Gemini analysis for skills, experience, education, and skill gaps

MiniGPT - Python, PyTorch, NumPy

GPT-style transformer language model built from scratch without external APIs or pretrained weights.

- Character tokenizer, causal self-attention, transformer blocks
- Training loop with loss tracking, checkpointing, and sampling

DevCode Online Compiler - Express.js, Monaco, Docker, TypeScript

Browser-based compiler with hardened Docker sandboxing and a Monaco editor.

- Network disabled, read-only rootfs, dropped capabilities
- CPU, memory, PID limits and hard wall-clock timeouts

MedAI Medical Query Generator - Flask, Gemini API, Python

Medical information assistant with safety-focused prompts and clear educational disclaimers.

- No diagnosis, no prescriptions, emergency escalation guidance
- Informational prototype only; not a clinical tool

NirajVersant - HTML, CSS, JavaScript, Web Speech API

English proficiency practice platform with 15 Versant-format tests and browser speech recognition.

- 15 tests, 70 questions each, practice scoring out of 80
- No accounts or tracking; runs entirely in the browser

Django E-Commerce - Django, Stripe, SQLite / PostgreSQL

E-commerce application with authentication, cart, checkout, Stripe payments, and order history.

- django-allauth authentication and address management
- Stripe Checkout integration and order records

EXPERIENCE

Software Engineer - HCLTech, Noida

Software engineering across programming fundamentals, databases, enterprise systems, cloud concepts, Git workflows, and professional skills.

Self-Directed Project Work - Independent (ongoing)

Built and published projects across AI, backend development, web technologies, and automation with an emphasis on inspectable source code and honest scope.

EDUCATION

Class 12 completed. School, board, grades, and dates are intentionally not listed.

ADDITIONAL PROJECTS

Bank Management System (Python, Tkinter, SQLite): role-based desktop app. All Tools In One Place (Next.js, TypeScript): utility platform with PDF, QR, formatting, encryption, and compression tools. Weather Alert Bot (FastAPI, React Native, Twilio): WhatsApp rain alerts. File Converter: deployed browser utility.